Schrödinger’s concerns were shown to be baseless, that there is no problem with reality in quantum physics because there is no need to think about reality in the first place.

Yet quantum physics is certainly telling us something about what is real, out in the world. Otherwise, why would it work at all? It would be very difficult to account for its wild success if it had no connection to anything real in the world. Even if the theory is simply a model, surely it’s modeling something and doing a reasonably good job of it. There must be some *thing* that ensures the predictions of quantum physics come to pass, with phenomenally high precision.

But figuring out what quantum physics is saying about the world has been hard. This is, in part, due to the sheer weirdness of the theory. Whatever is in the world of the quantum, it is nothing familiar at all. The seemingly contradictory nature of quantum objects—atoms that are here and there at the same time, radiation that has both been emitted and remains latent in its source—isn’t the only alien aspect of the theory. There are also instantaneous long-distance connections between objects: subtle, useless for direct communication, but surprisingly useful for computation and encryption. And there does not appear to be any limit to the size of object that is subject to quantum physics. Ingenious devices built by experimental physicists coax larger and larger objects to display strange quantum phenomena almost monthly—deepening the gravity of the problem that no such quantum phenomena are seen in our everyday lives.

These phenomena aren’t the only challenge to deciphering the message of quantum physics. They’re not even the largest challenge. Despite the fact that every physicist agrees that quantum physics works, a bitter debate has raged over its meaning for the past ninety years, since the theory was first developed. And one position in that debate—held by the majority of physicists and purportedly by Bohr—has continually denied the very terms of the debate itself. These physicists claim that it is somehow inappropriate or unscientific to ask what is going on in the quantum realm, despite the phenomenal success of the theory. To them, the theory needs no interpretation, because the things that the theory describes aren’t truly real. Indeed, the strangeness of quantum phenomena has led some prominent physicists to state flatly that there is no alternative, that quantum physics proves that small objects simply do not exist in the same objectively real